

Image Processing & Recognition: Lecture 8 Review

Welcome to Level 8! We are now combining **Vision** (seeing) with **Language** (speaking). This is where AI starts to feel truly intelligent.

Part 1: What are you learning? (The "Human" Summary)

1. The Core Task: Image Captioning

- **Goal:** Input an image $\rightarrow$ Output a sentence describing it (e.g., "A dog catching a frisbee").
- **Mechanism:** It connects a **CNN** (which "sees" the image) to an **RNN/LSTM** (which "writes" the sentence).

2. The Models

- **NIC (Neural Image Caption):** A simple Encoder-Decoder.
 - **Encoder:** A pre-trained CNN (like VGG) takes the image and turns it into a single vector (feature map).
 - **Decoder:** An LSTM takes that vector as the *first* input (at $t = -1$) and then starts predicting words one by one.
- **LRCN (Long-term Recurrent Convolutional Networks):** Similar to NIC, but more flexible. It can handle video (sequences of images) or inject the image features at every time step of the LSTM, not just the first one.

3. Adding Attention

- **The Problem:** Compressing an entire complex image into one vector is hard.
- **The Solution (Visual Attention):** Instead of looking at the whole image at once, the LSTM "looks" at specific regions of the image for each word it generates.
 - When generating "dog," it looks at the dog pixels.
 - When generating "frisbee," it shifts focus to the frisbee pixels.

4. Visual Question Answering (VQA)

- **Goal:** Input an image + a question ("How many horses?") $\rightarrow$ Output an answer ("2").
- **Mechanism:** Two separate neural networks process the inputs:
 1. **Image Channel:** CNN processes the picture.
 2. **Question Channel:** LSTM processes the text question.
 3. **Fusion:** The two vectors are combined (e.g., multiplied element-wise) and sent to a classifier to pick the best answer.

Part 2: The Generated Solved Problem (Step-by-Step)

Since the PDF did not have a specific exercise, I have designed one based on the core mechanism of **Visual Question Answering (VQA)**: fusing image and text features.

Exercise 8

We want to predict an answer in a VQA system by fusing image and question features via **element-wise multiplication**.

Given:

- **Image Embedding (v):** A vector representing the image features (e.g., from a VGG Net).

$$v = [0.5, 1.0, -0.5]$$

- **Question Embedding (q):** A vector representing the question features (e.g., from an LSTM).

$$q = [1.0, 0.5, 2.0]$$

- **Classifier Weights (W):** A matrix mapping the fused vector to 2 possible answers ("Yes", "No").

$$W = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix}$$

(Row 1 is for "Yes", Row 2 is for "No")

- **Bias (b):**

$$b = [0, 0]$$

Task:

1. Calculate the **Fused Vector (h)**.
2. Calculate the **Logits** (raw scores) for "Yes" and "No".
3. Determine the predicted answer.

Solution:

Step 1: Fuse the Embeddings The lecture describes fusing via element-wise multiplication (Hadamard product).

- Formula: $h = v \odot q$
- **Source: Page 43** mentions: "The transformed image and LSTM embeddings... are then fused via element-wise multiplication."

$$h = [0.5 \times 1.0, \quad 1.0 \times 0.5, \quad -0.5 \times 2.0]$$

$$h = [0.5, \quad 0.5, \quad -1.0]$$

Step 2: Calculate Logits Pass the fused vector through the fully connected layer.

- Formula: $z = W \cdot h + b$
- **Source: Page 43** mentions: "passed to an MLP... followed by a softmax layer."
- **Score for "Yes" (Row 1):**

$$z_{yes} = (1 \times 0.5) + (0 \times 0.5) + (1 \times -1.0) = 0.5 + 0 - 1.0 = -0.5$$

- **Score for "No" (Row 2):**

$$z_{no} = (0 \times 0.5) + (1 \times 0.5) + (-1 \times -1.0) = 0 + 0.5 + 1.0 = 1.5$$

Step 3: Prediction The model outputs the class with the highest score.

- $z_{yes} = -0.5$
- $z_{no} = 1.5$

Result: The predicted answer is **"No"**.

Part 3: Your Turn (Interactive Exercises)

Practice Problem 1: Image Captioning (Unrolled LSTM)

Let's simulate one step of an LSTM generating a caption.

Setup:

- **Current Hidden State (h_{t-1}):** $[1, 1]$.
- **Current Input Word Vector (x_t):** $[0, 2]$. (e.g., the word "dog")
- **Weights:**
 - W_{hx} (Input-to-Hidden): Identity matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
 - W_{hh} (Hidden-to-Hidden): Identity matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.
- **Activation:** Standard RNN tanh. $\tanh(x)$.

Task: Calculate the next hidden state h_t .

Solving Explanation:

1. **Formula:** $h_t = \tanh(W_{hx}x_t + W_{hh}h_{t-1})$.
2. **Input Term:** $[1, 0; 0, 1] \times [0, 2] = [0, 2]$.
3. **Hidden Term:** $[1, 0; 0, 1] \times [1, 1] = [1, 1]$.

4. **Sum:** $[0, 2] + [1, 1] = [1, 3]$.
5. **Activation:** $h_t = [\tanh(1), \tanh(3)] \approx [0.76, 0.99]$.

Practice Problem 2: Parameter Counting (VQA Fusion)

Setup:

- **Image Embedding Size:** 1024.
- **Question Embedding Size:** 1024.
- **Fusion Type:** Element-wise multiplication.
- **Output Classifier:** Fully connected layer mapping 1024 features to 1000 answers.

Task: Calculate the number of parameters in the **Output Classifier** layer (Weights + Biases).

Solving Explanation:

1. **Input Size:** 1024 (Since element-wise multiplication doesn't change the vector size, the fused vector is still 1024).
2. **Output Size:** 1000.
3. **Weights:** $1024 \times 1000 = 1,024,000$.
4. **Biases:** 1000.
5. **Total:** 1,025,000 parameters.